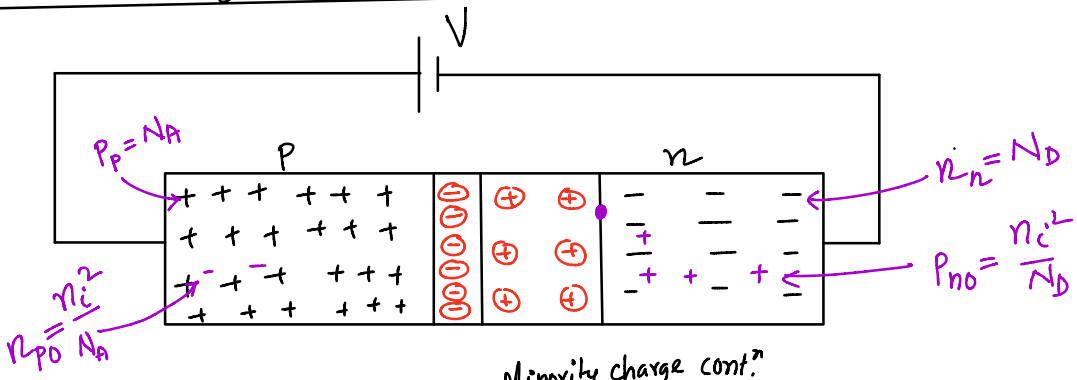
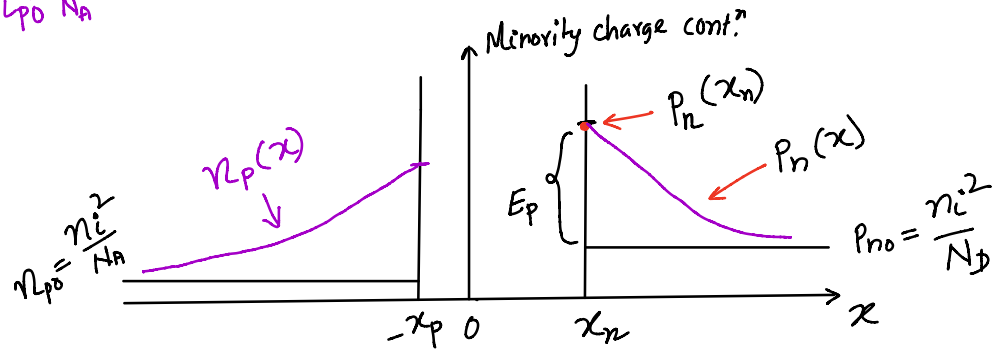
Derivation of Diode Characteristic Equation





$$V(x_n) - V(-x_p) = V_T \ln\left(\frac{p_p(-x_p)}{p_n(x_n)}\right)$$

$$\Rightarrow V_0 - V = V_T \ln\left(\frac{N_A}{p_n(x_n)}\right)$$

$$\Rightarrow \frac{p_n(x_n)}{N_A} = e^{\frac{V - V_0}{V_T}}$$

$$\Rightarrow p_n(x_n) = \underbrace{N_A}\, e^{V/V_T} \cdot \underbrace{e^{-\frac{V_0}{V_T}}} \longrightarrow (i)$$

We know, $$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$\Rightarrow -V_0 = V_T \ln\left(\underbrace{\frac{n_i^2}{N_D}}_{P_{n0}} \cdot \frac{1}{N_A}\right)$$

$$\Rightarrow P_{n0} = N_A e^{-V_0/V_T} \longrightarrow \text{(ii)}$$

From (i) & (ii), we get

$$\boxed{P_n(x_n) = P_{n0}\, e^{V/V_T}}$$

So, the excess minority carrier concentration is

$$E_p = P_{n0} e^{V/V_T} - P_{n0}$$

$$= P_{n0}\left(e^{V/V_T} - 1\right)$$

This excess minority carrier will decay exp. and we can write $P_n(x)$ as

$$P_n(x) = \underline{P_{n0}} + E_p e^{-(x-x_n)/L_p}$$

where L_p is a constant called diffusion length of the holes in n-type material.

So, this profile will lead to a diffusion current

$$J_p(x) = -q D_p \frac{d P_n(x)}{dx}$$

$$\Rightarrow J_p(x) = \frac{q D_p E_p}{L_p} e^{-(x-x_n)/L_p}$$

$$\therefore J_p(x) = q\left(\frac{D_p}{L_p}\right) P_{no}\left(e^{V/V_T}-1\right) e^{-(x-x_n)/L_p}$$

So, $J_p(x)$ is max at $x = x_n$

$$\therefore J_p(x_n) = q\left(\frac{D_p}{L_p}\right) P_{no}\left(e^{V/V_T}-1\right)$$

Similarly, on the p-side we will get

$$J_n(-x_p) = q\left(\frac{D_n}{L_n}\right) n_{po}\left(e^{V/V_T}-1\right)$$

$$\therefore J = J_n(x_n) + J_n(-x_p)$$

$$\therefore I = A J$$

$$= Aq\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)\left(e^{V/V_T}-1\right)$$

$$\because\ p_{n0} = \frac{n_i^2}{N_D} \quad \text{and} \quad n_{p0} = \frac{n_i^2}{N_A}$$

$$\boxed{\therefore\ I = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left(e^{V/V_T}-1\right)}$$

$$\boxed{I = I_s\left(e^{V/V_T}-1\right)}$$

where $I_s = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$